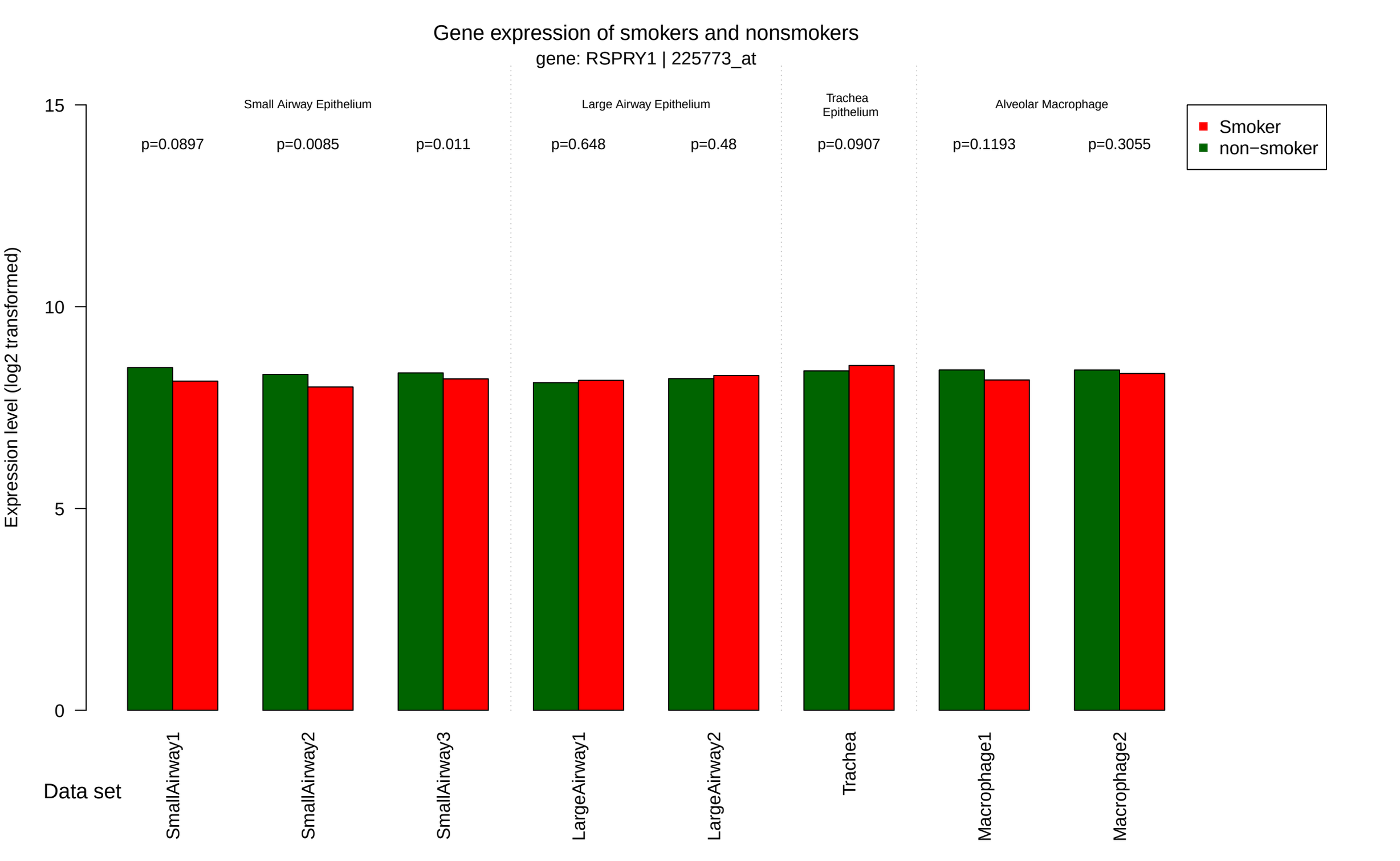
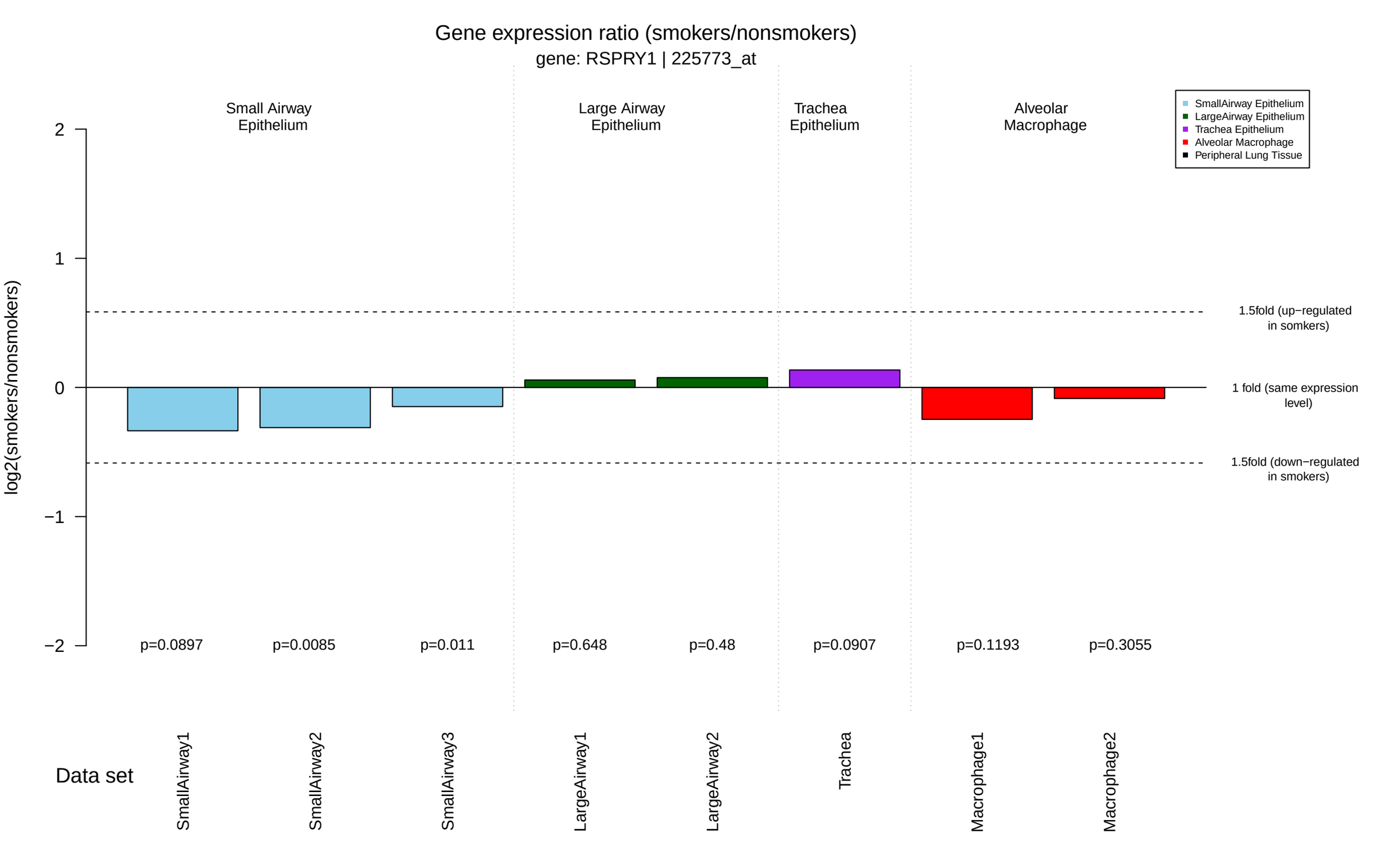
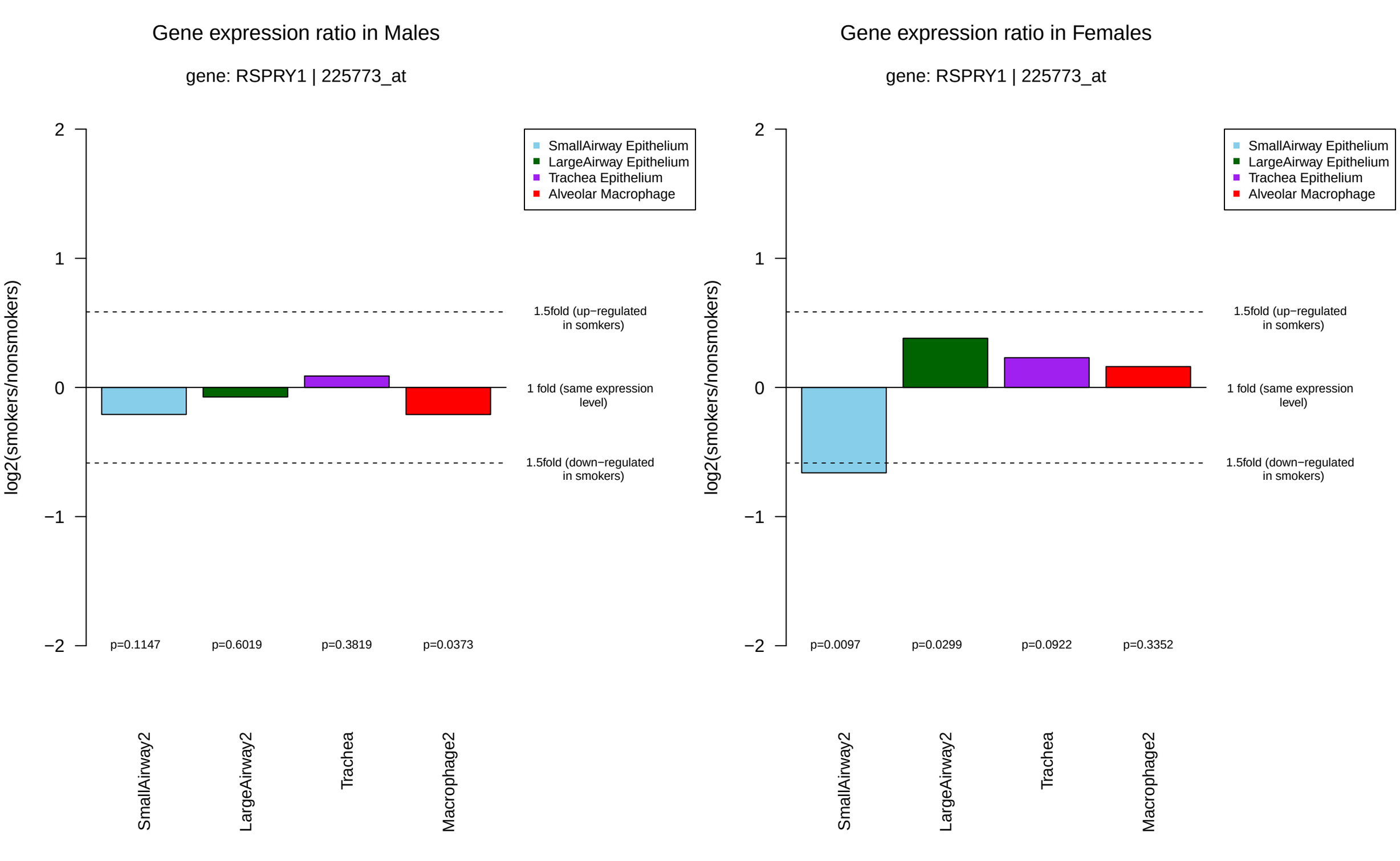
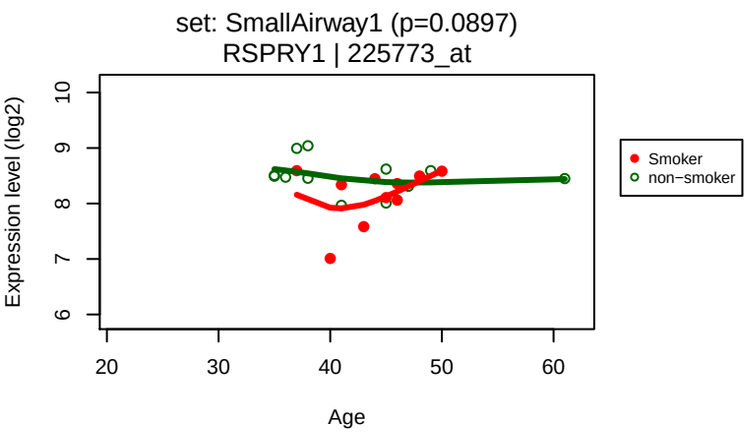
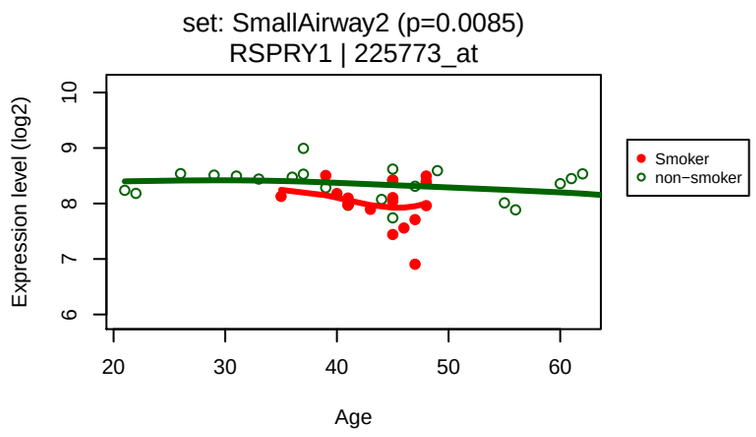
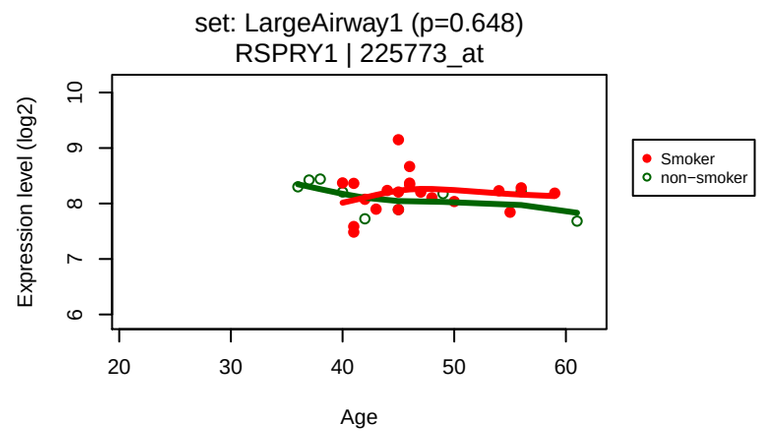

Gene expression of smokers and nonsmokers
gene: RSPRY1 | 225773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0897
p=0.0085
p=0.011
p=0.648
p=0.48
p=0.0907
p=0.1193
p=0.3055
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

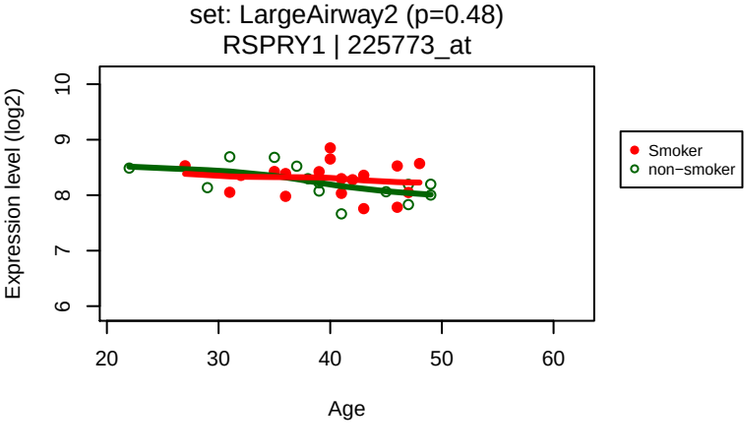

Gene expression ratio (smokers/nonsmokers)
gene: RSPRY1 | 225773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0897
p=0.0085
p=0.011
p=0.648
p=0.48
p=0.0907
p=0.1193
p=0.3055
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

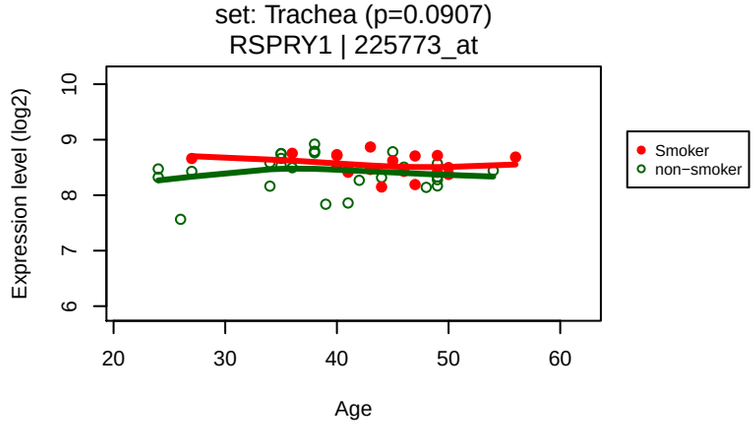

Gene expression ratio in Males
gene: RSPRY1 | 225773_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1147
p=0.6019
p=0.3819
p=0.0373
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSPRY1 | 225773_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0097
p=0.0299
p=0.0922
p=0.3352
SmallAirway2
LargeAirway2
Trachea
Macrophage2

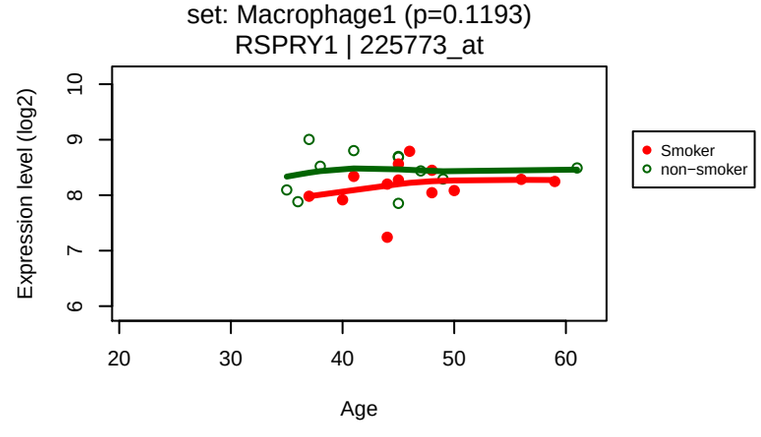

set: SmallAirway1 (p=0.0897)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker

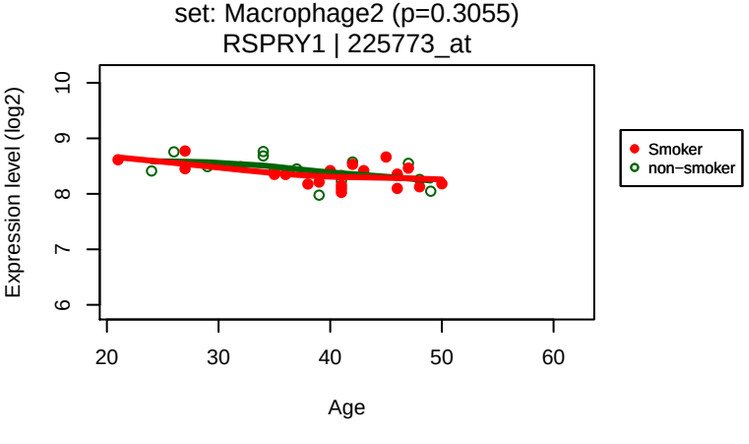

set: SmallAirway2 (p=0.0085)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker

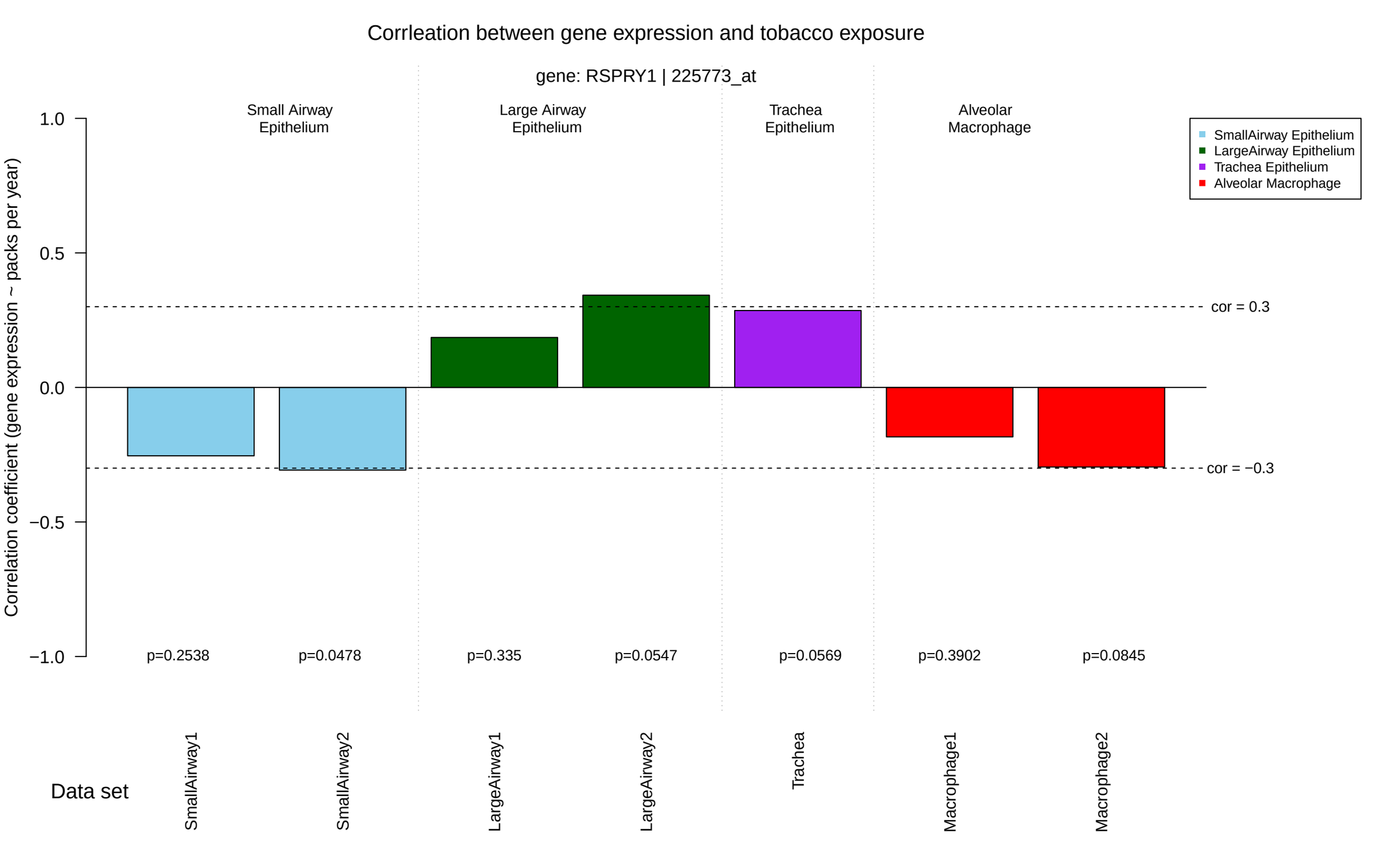

set: LargeAirway1 (p=0.648)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.48)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0907)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1193)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3055)
RSPRY1 | 225773_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSPRY1 | 225773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2538
p=0.0478
p=0.335
p=0.0547
p=0.0569
p=0.3902
p=0.0845
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2